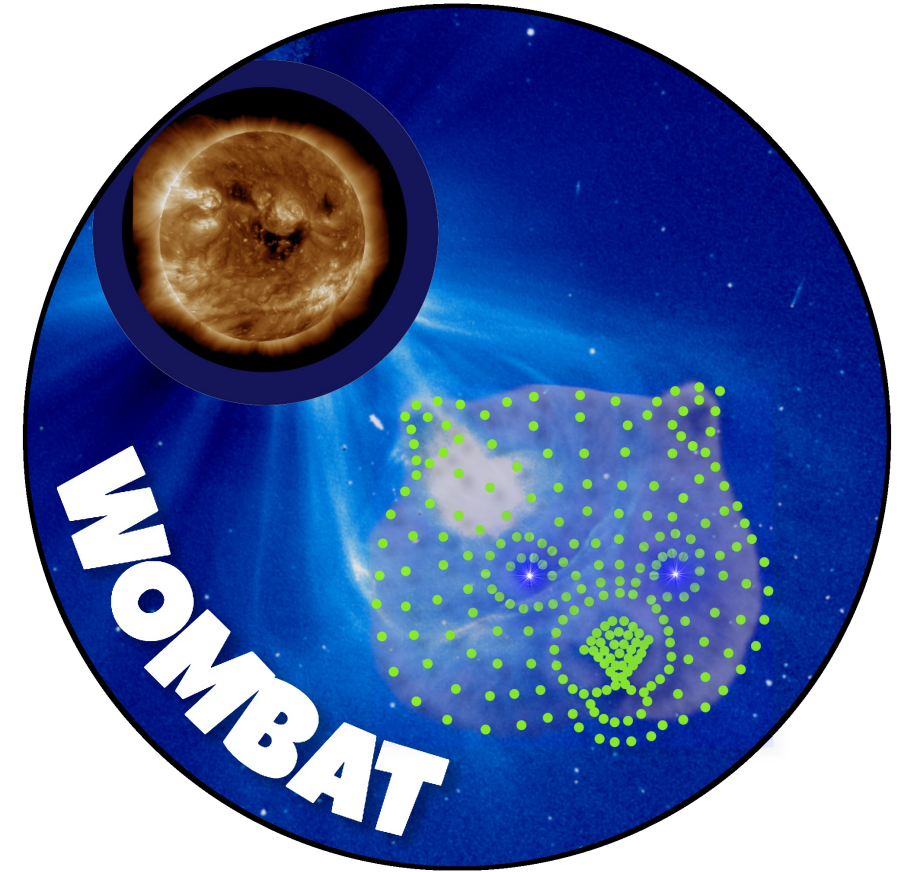


Using WOMBAT and DINGO



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Initial Download and Set Up

1. Download WOMBAT from [Github repository](#)
2. Download support files from [Zenodo repository](#)
3. Make sure Python has the libraries it needs

No need to install/compile the code itself

Library Option 1 - Using PIP

```
pip install -r requirements.txt
```

Library Option 2 - Using Conda

Set up conda (good instructions here)

Create environment

```
conda env create -f wombat.yml --name wombat
```

Initiate environment

```
conda activate wombat
```

Deactive environment (when done)

```
conda deactivate
```

Required Packages

```
aiapy 0.10.2, astropy 7.1.1,  
numpy 2.3.4, pyqt5 5.15.11,  
pyqtgraph 0.13.7, scipy 1.16.3,  
spiceypy 8.0.0, sunpy 7.0.3,  
wget 3.2
```

Also uses: datetime, itertools,
math, os, platform, and sys

Getting Observations – WOMBAT Pull

Basic command:

```
python wombatPullObs.py t1 t2 insts timeRes
```

Time Arguments (Required)

Start/end time as YYYY-MM-DDTHH:MM
(or any form accepted by Sunpy parse_time)

Time Resolution (Optional)

Set time resolution by observation type:

- EUV images - **EUVT#**
- Coronagraphs - **CORT#**
- Heliospheric imagers - **HIT#**
(# in minutes, defaults 10/20/30)

Satellite	Instrument Tags (1+ required)
PSP	WISPR, WISRPI, WISPRO
SDO	AIA94, AIA131, AIA171, AIA193, AIA211, AIA304, AIA335, AIA1600, AIA1700
SOHO LASCO	C2, C3
Solo	SoloHI, SoloHI1, SoloHI2, SoloHI3, SoloHI4
STEREO	EUVI171, EUVI195, EUVI284, EUVI304, COR1, COR2, HI1, HI2 <i>*Can specify A or B on the end</i>

Uses Sunpy Fido/wget to pull data from the web, stores in **pullFolder/**

Getting Observations – Not Fido

There is no requirement to use WOMBAT pull to obtain the files, but it is a convenient option when Sunpy Fido plays nice. It is sensitive to network configurations and doesn't always work (particularly for WISPR/SoloHI).

As long as the right level files are used (and have the same naming conventions as what Fido pulls) you can just copy fits files into the pullFolder in the appropriate spots

File structure is wombat/pullFolder/satellite/instrument/*wavelength or detector if needed*)

e.g. wombat/pullFolder/PSP/WISPR/Inner

wombat/pullFolder/STEREO/EUVIA/171

Prepping Observations – WOMBAT Process

Basic command:

```
python wombatProcessObs.py t1 t2 insts name.pkl
```

Prep Components

Find images in time range for selected insts
Standard image processing (solarsoft ports)
Running/base differences
Determine mass image from total brightness
Convert to nice ranges on different scales
Save pickle (binary python file) in **wbPickles/**

Arguments

t1, t2, insts - same as pull call (required)
name.pkl – a name for the output pickle
(defaults to *WBGUI_temp.pkl*)

***Note SR and LW tags will calc mass from corresponding L2 data (if it can find it)

Inst	Input file level	Processing Notes
AIA	L1 (aia.lev1..fits)	Uses aiapy
C2/C3	L0.5 (img#.fits)	SSW port
COR1	Tags sc4X, s5xX, n4cX (X = A or B)	SSW port. Requires download of monthly_min backgrounds ***
COR2	Tags d4c2X	SSW port. Total brightness or pB triplets
EUI	L2 (FSI only)	Uses basic sunpy map
EUVI	n4euX, n5euX	SSW port
HI1/HI2	s4hi1X, s4h2X	SSW port. Can add tag “_SR” for star removal processing
SoloHI	L2 (1ft, 2ft, 3ft/g, 4t/g)	SSW port. Makes mosaic if not given 1-4 tag
WISPR	L3 (1221, 2222)	SSW port. Can add tag “_LW” or “_L2” if have pulled those forms of imgs

Launching WOMBAT

Basic command:

```
python wombatWrapper.py pickleName nWFs ovw
```

Reload command:

```
python wombatWrapper.py logFile id1+id2... ovw
```

Basic Arguments

pickleName.pkl – the output of the processing script (default is wbPickles/WBGUI_temp.pkl)

nWFs – the number of wireframes to use (max 5)

ovw – flag to include the overview window (defaults to off)

Reload Arguments

logFile – a log file from WOMBAT that is used to reload wireframe(s) (default in wbOutputs/WomBlog.txt)

ids – a string of integer ids to indicate line numbers in the logFile with values separated by + (e.g. 5+6 is lines 5 & 6)

Parameter Window (PW)

Basic launch opens the parameter window with all widgets unset/at default values

1. *Wireframe (WF) tab selection*
2. *Time slider*
3. *WF color selection*
4. *WF type selection*
5. *Log current WF parameters button*
6. *Show/hide WF button*
7. *Output name text box (for logFile/reload)*
8. *Background difference selection*
9. *Background scaling selection*
10. *Save figure button*
11. *Calculate mass button*
12. *Exit button*



WF Types

GCS
Torus
Sphere/Half Sphere
Ellipse/Half Ellipse
Slab
Tube
GCS* (movable origin)

Difference Types

Running*
Base
*Can replace with
SREM style RD

Background Scaling

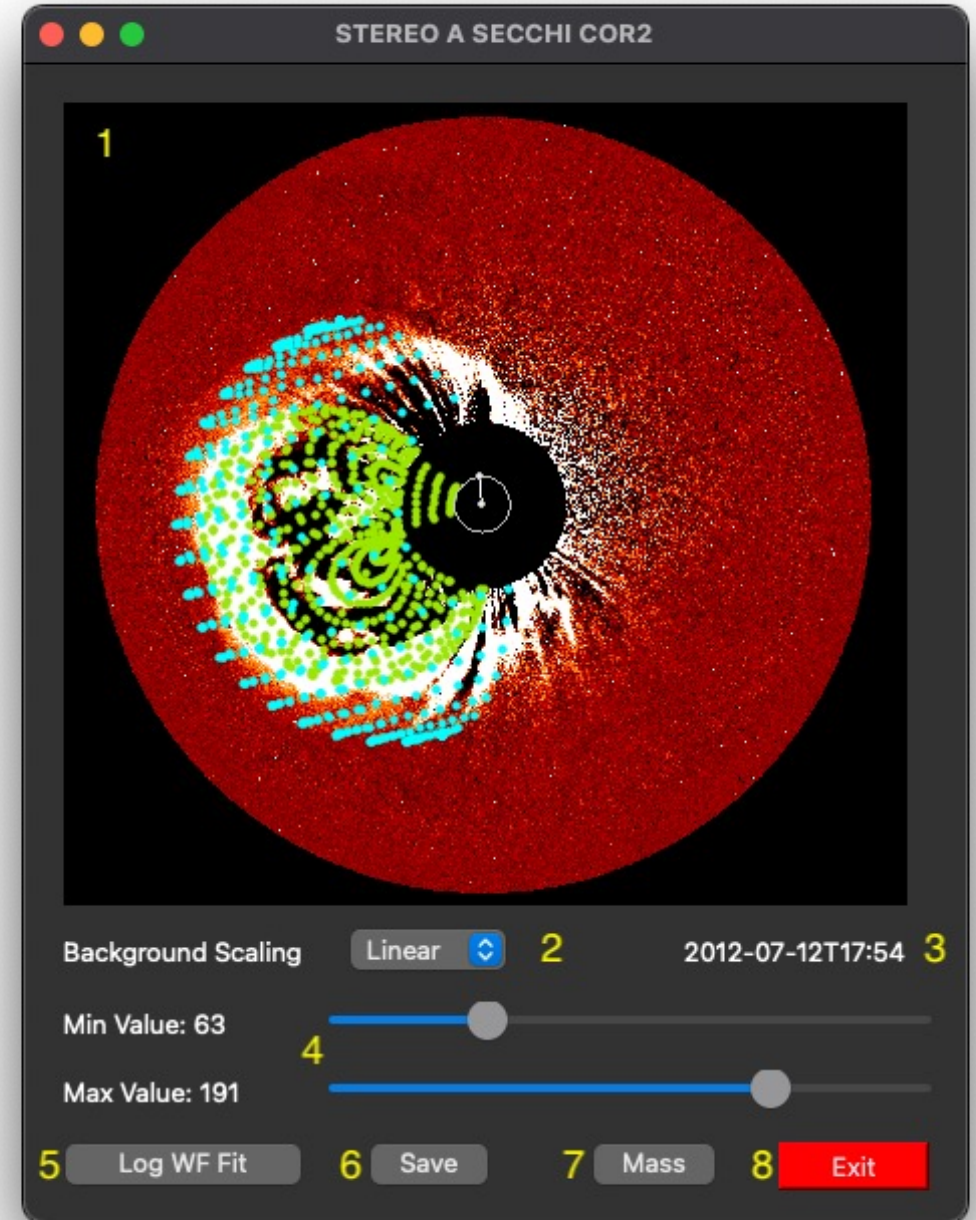
Linear
Logarithmic
Square root

Observation Window (OW)

Each instrument saved in the pickle will have its own observation window

1. *Display window (with instrument and time labels)*
2. *Background scaling selection*
3. *Observation time stamp*
4. *Min/max plot range sliders*
5. *Log current WF parameters butto*
6. *Save button*
7. *Mass button*
8. *Exit button*

The widgets matching the parameter window ones will perform the same action but only for this instrument (as opposed to all)

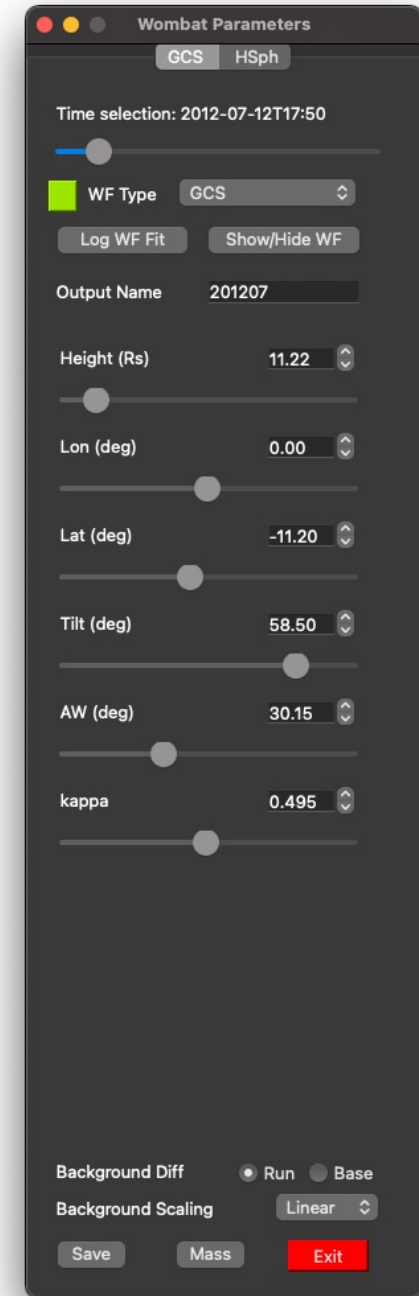


Setting A WF Type

Selecting a WF type causes parameters to appear in the PW and a WF to appear in the OW.

Adjust sliders to get visual match.

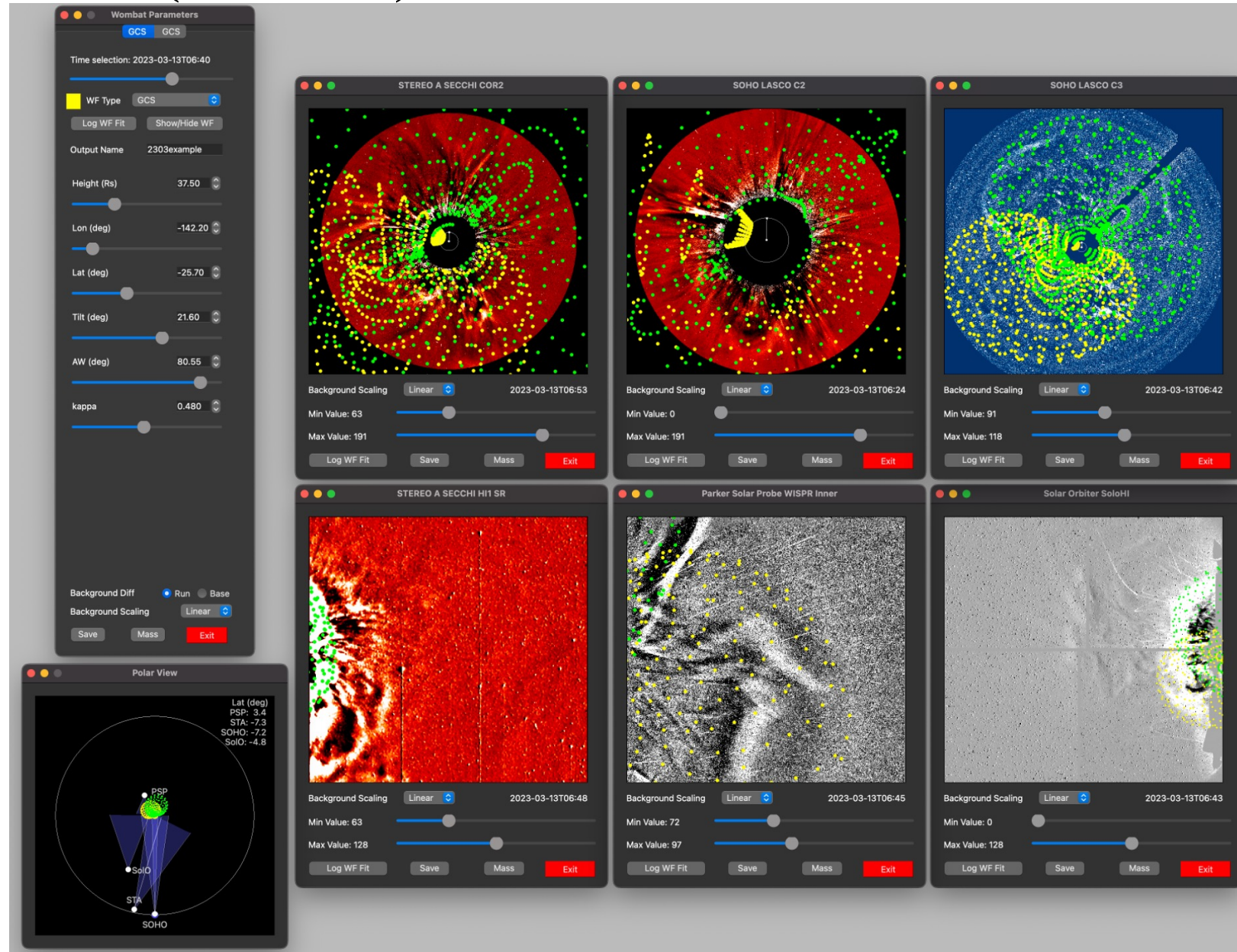
Can fine tune values with the arrows or using the text boxes. Must press return in text box for adjustment to take place



Overview Window (OVW) and More

Example showing two WFs on six different observations and the overview window

OVW shows a top down view each satellite and their FoV and the WF projections

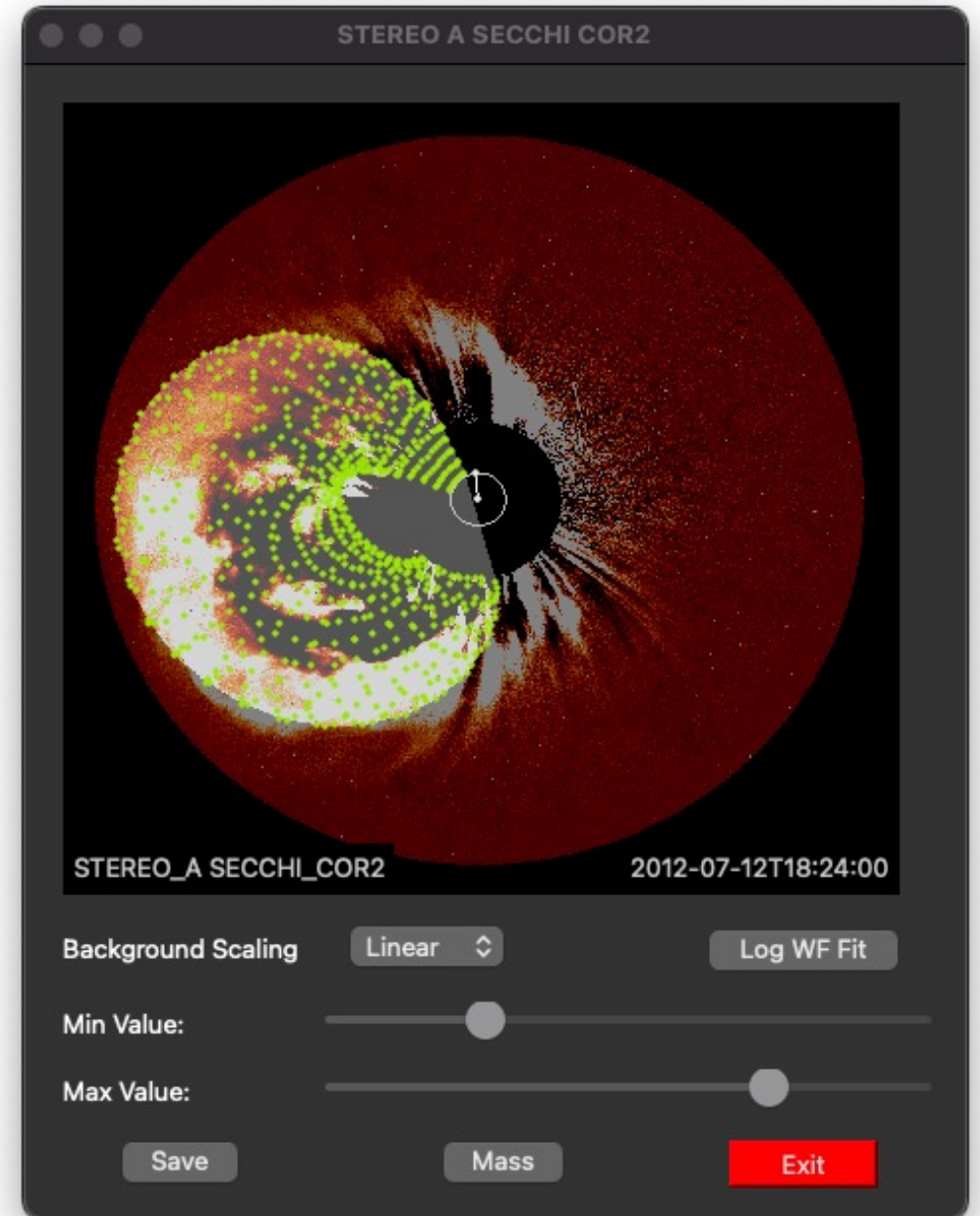


Mass Calculation

Pressing the mass button will create a mask surrounding the WF shape, sum up the total mass in that region, and print the results to the command line.

Pressing the button a second time will remove the mask.

It will **work** with multiple WFs by running each individually as if the others do not exist. This overestimate the total mass where the WFs overlap. *DINGO can do better for these cases.*



Logging, Saving, and Shortcuts

Logging a result adds information needed to reconstruct a WF to the log file

Time of logging, inst, obs time, WF type, 9 WF params ("None" if unused), background pickle, time ID within pickle

Saving saves images of each OW (and OVW)

Figs saved as wombat_datetime_inst.png in wbOutputs/

The PW text box can be used to set the name of these files

Defaults to WomBlog.txt and wombatSummaryFile.txt, both in wbOutputs/

(⇧) indicates that pressing that just the key will perform the action for the current time but shift+key will do it for all times. The entries without (⇧) around ⇧ only operate when shift is pressed. These actions work on a single obs window if it is selected or on all the obs windows when the param is window is active

WOMBAT keyboard shortcuts

Key	Action
Q	Quit current window
ESC	Quit everything
H	Show/hide wireframes
←	Move to earlier times
→	Move to later times
(⇧)L	Log current reconstruction
(⇧)S	Save figs
(⇧)M	Mass calculation
(⇧)R	Running difference mode
(⇧)B	Base difference mode
(⇧)1	Linear background
(⇧)2	Logarithmic background
(⇧)3	Square root background
⇧7	Copy WF params back in time
⇧8	Copy WF params forward in time
⇧9	Copy plot min/max back in time
⇧0	Copy plot min/max forward in time

Running wombatPlots

python wombatPlots.py *logFile ids mode bonusParams*

Basic Arguments

logFile and **ids** – same as reloading GUI

Mode = type + #params

Types:

- **ht** – just height
- **kin** – height + vel + accel (kinematics)
- **en** – kin + mass + KE (energetics)

Param #:

- **1** – just height
- **2** – height + AWs
- **3** – all WF params

e.g. kin2 is height, vel, acc, and AWs

Bonus Parameters

Flags

eb – include error bars
drag – fit drag eq
vsh – plot versus height
log – plot x-axis heights on log scale
png/pdf – set image type
1au/L1 – get pred arrival time at that point

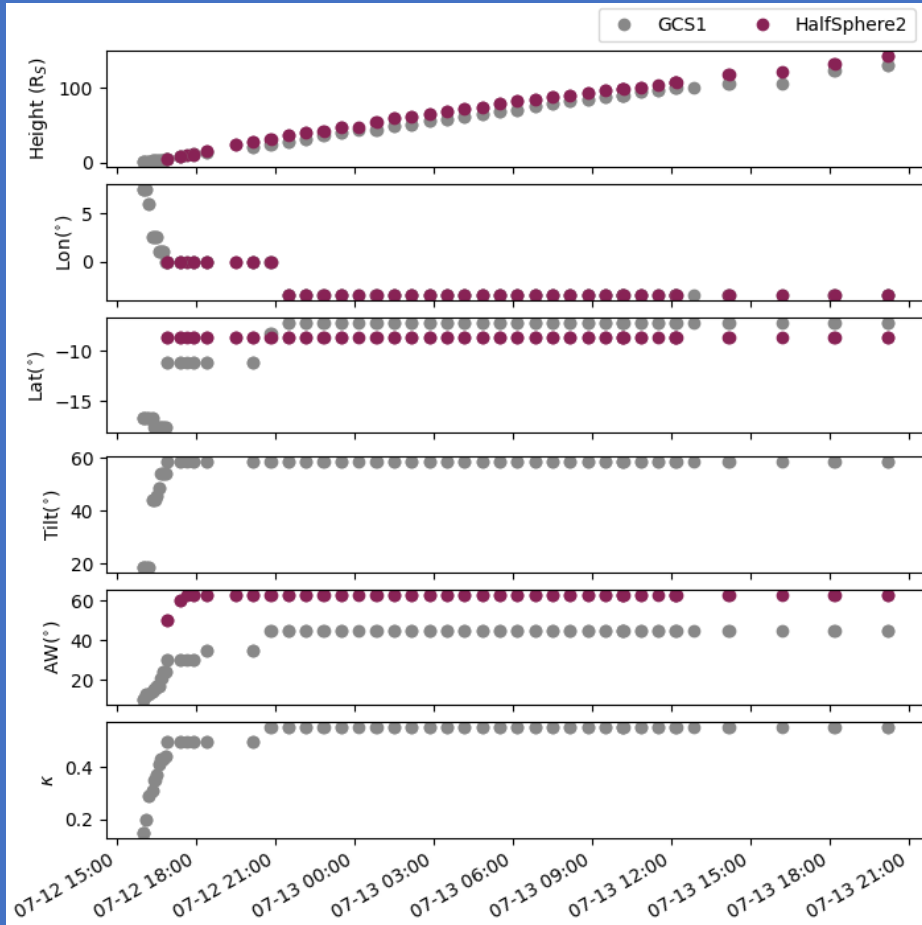
Prefix_# options – sets prefix at #

dh1_#/dh2_# – set height range to check on drag model
densratio_# – density ratio between inner/outer WF
newbase_# – a time stamp to use for a new base diff/masses

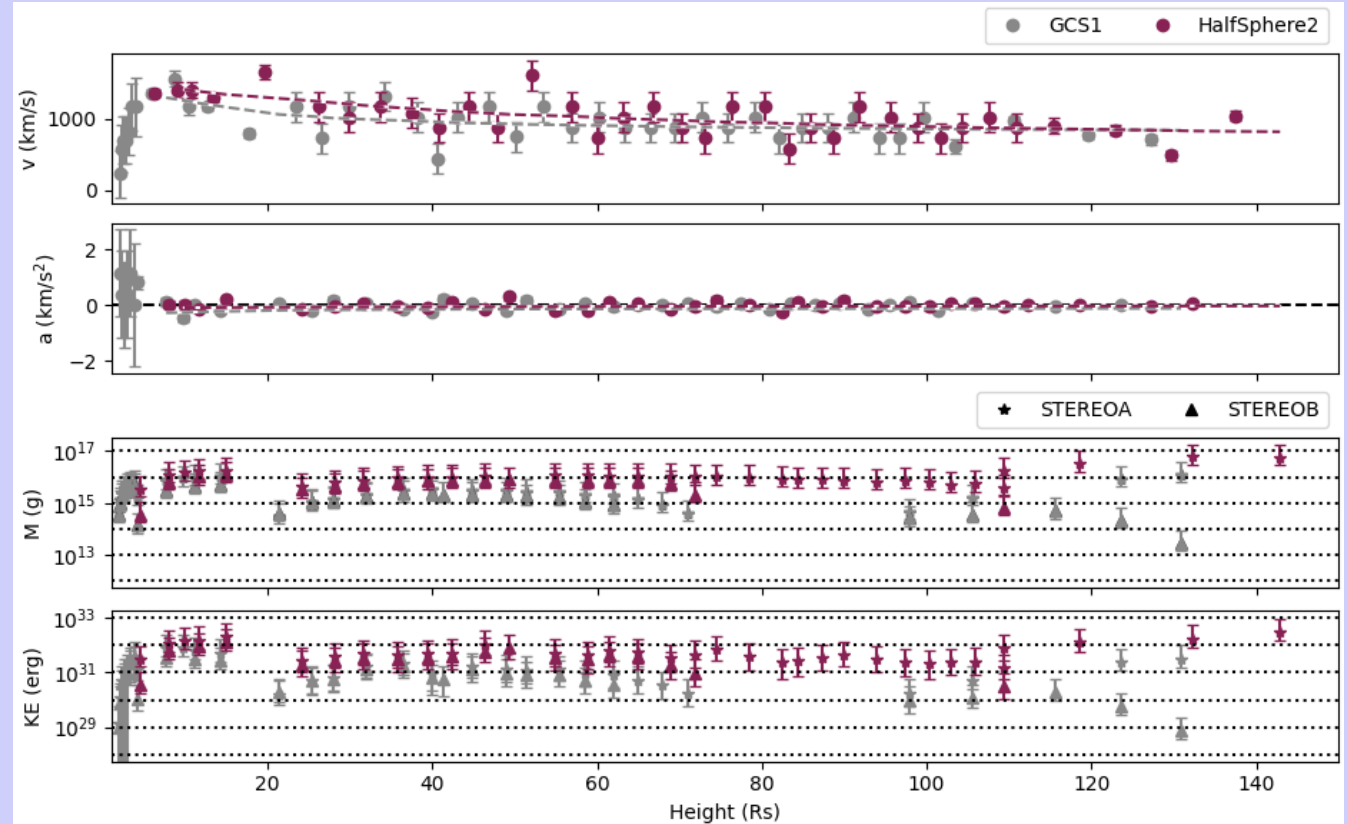
saveName – name for saving figure (will pop up window otherwise)

pickleName – name of a pkl from a previous plot run*

wombatPlots Examples



python wombatPlots.py logFile ids ht3



python wombatPlots.py logFile ids en1 vsr eb drag

Other wombatPlots Options

```
wombatPoints.txt — wombat (git: main)
1 GCS1 2012-07-12T16:20 0 1.046 -0.006 0.108
2 GCS1 2012-07-12T16:20 1 1.043 0.024 0.135
3 GCS1 2012-07-12T16:20 2 1.038 0.057 0.159
4 GCS1 2012-07-12T16:20 3 1.032 0.093 0.178
5 GCS1 2012-07-12T16:20 4 1.026 0.130 0.193
6 GCS1 2012-07-12T16:20 5 1.018 0.169 0.202
7 GCS1 2012-07-12T16:20 6 1.010 0.209 0.207
8 GCS1 2012-07-12T16:20 7 1.001 0.249 0.207
9 GCS1 2012-07-12T16:20 8 0.991 0.289 0.201
10 GCS1 2012-07-12T16:20 9 0.981 0.327 0.191
11 GCS1 2012-07-12T16:20 10 0.971 0.364 0.175
12 GCS1 2012-07-12T16:20 11 0.961 0.398 0.155
13 GCS1 2012-07-12T16:20 12 0.951 0.429 0.131
14 GCS1 2012-07-12T16:20 13 0.942 0.457 0.103
15 GCS1 2012-07-12T16:20 14 0.932 0.482 0.071
16 GCS1 2012-07-12T16:20 15 0.924 0.502 0.037
17 GCS1 2012-07-12T16:20 16 0.916 0.517 -0.000
18 GCS1 2012-07-12T16:20 17 0.909 0.528 -0.039
19 GCS1 2012-07-12T16:20 18 0.902 0.534 -0.079
20 GCS1 2012-07-12T16:20 19 0.897 0.535 -0.120
21 GCS1 2012-07-12T16:20 20 0.893 0.531 -0.160
22 GCS1 2012-07-12T16:20 21 0.891 0.523 -0.200
23 GCS1 2012-07-12T16:20 22 0.889 0.509 -0.239
24 GCS1 2012-07-12T16:20 23 0.889 0.490 -0.275
25 GCS1 2012-07-12T16:20 24 0.890 0.468 -0.310
26 GCS1 2012-07-12T16:20 25 0.892 0.441 -0.341
27 GCS1 2012-07-12T16:20 26 0.895 0.411 -0.368
```

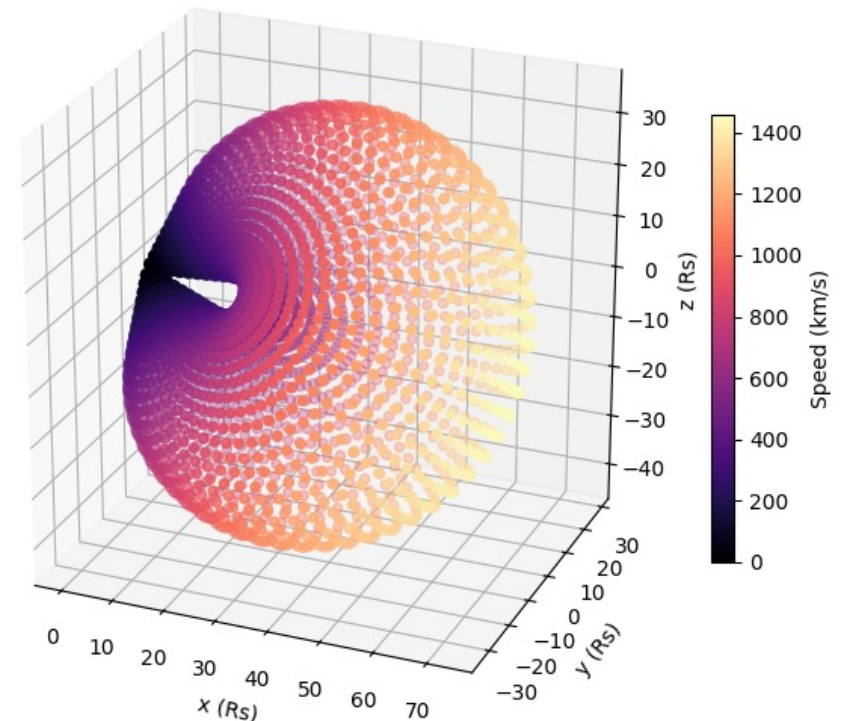
Generate list of wireframe points for each recon

```
python wombatPlots.py logFile ids pts
```

File has WFname obsTime pointNumber x y z (pts as Rsun in Stony Cart)

Map of velocities on WF surface from two recons at different times

```
python wombatPlots.py logFile id1+id2 vmap
```



*Will save points and velocities in same format as pts with vx vy vz added (in km/s) if given a save name

Running wombatMovie

python wombatMovie.py

There are too many options to configure via command line so the top portion of wombatMovie.py needs to be edited before it is run. This section is labeled “Configure here!” and stops at “No touching!”

The parameter definitions are given in the comments within the file

```
19 #|-----|
20 #|-----|
21 #|----- Configure here!!! -----|
22 #|-----|
23 #|-----|
24 global doClean, customOrder, ovw, doCutout, cutouts
25
26 # Name of wombat log file
27 logFilePath = 'wbOutputs/202303full.txt'
28
29 # Movie save name. Should end with .mp4 (other formats untested)
30 movieName = 'temp.mp4'
31
32 # Lines to do, same string format as other wombat functions
33 idstr = '1-423'
34
35 # Time Resolution (in minutes)
36 tRes = 30
37
38 # Number of columns in movie (max 5)
39 nHoriz = 4
40
41 # Include clean imgs without wf proj
42 doClean = True
43
44 # Frames per second
45 fps = 4
46
47 # Flag to set the instrument order
48 # (must include all the inst in the pickle)
49 customOrder = True
50 instOrder = ['C2', 'COR2A', 'C3', 'WISPRI', 'SoloHI', 'HI1A_SR']
51 #instOrder = ['COR1B', 'COR1A', 'COR2B', 'COR2A']
52
53 # Running (0) or base diff (1)
54 didx = 0
55 # Scaling mode linear(0), log(1), or sqrt(2)
56 sclidx = 0
57 # Wireframe scatter point size
58 wfSize = 3
59
60 # Option to include overview plot
61 ovw = True
62
```

Running DINGO

python dingo.py *logFile ids dim bonusParams*

Basic Arguments

logFile and **ids** – same as reloading GUI

Dimensions (dim)

OD – calculate total mass in WF projection

1D – line profiles of in place + in situ density at a target satellite

2D – contours of density

23D – contours of density shown in 3D

3D – scatter plot of a cloud of densities

Bonus Parameters

Flags

projoff – no projection effects (proj. effects included by default)

logplot – plot 2D contours on log scale instead of linear

Prefix_# options – sets prefix at #

expf1_#, expf2_#, vcme_# – set 1D propagation model

densratio_# – density ratio between inner/outer WF

newbase_# – a time stamp to use for a new base diff/masses

target – in situ satellite name, required for 1D mode (use one string from ACE, BepiColombo, DSCOVR, MAVEN, PSP, SoLo, STEREO-A, STEREO-B, VEX, or Wind)

saveName – string to add to output figure/file names

DINGO 0D + 3D

The simple modes – 0D prints the total mass to the terminal and 3D opens an interactive scatter plot

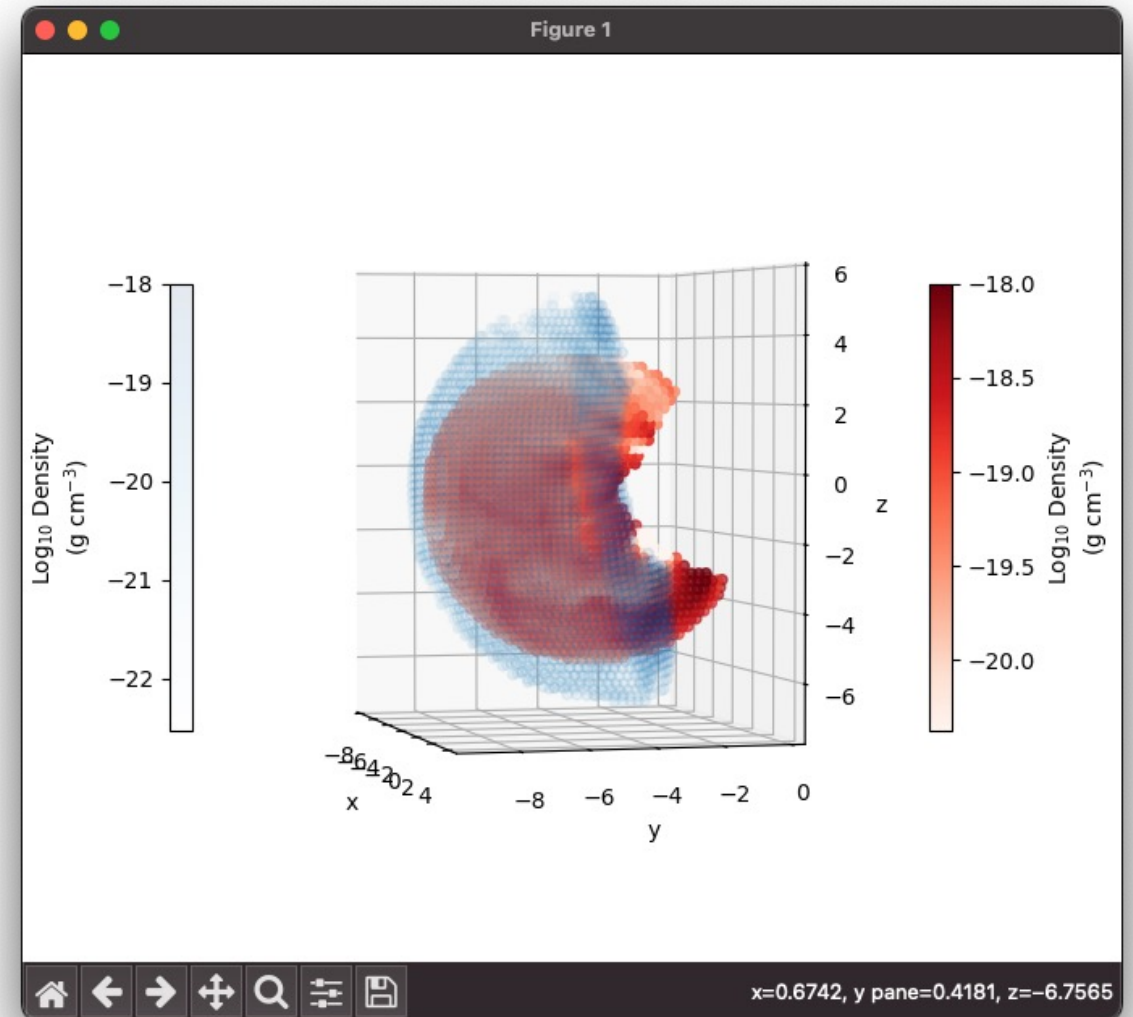
DINGO 0D

Example results for 2 WFs and 2 time steps.

Lists time, instrument, WF1 type, WF2 type, Mass1, Mass2

Time	Inst	WF1	WF2	Mass1(1e15 g)	Mass2(1e15 g)
2012-07-12T17:39:00	COR2A	GCS	HalfSphere	7.97	14.25
2012-07-12T18:24:00	COR2A	GCS	HalfSphere	10.91	20.07

Can use built in Python window features to resize, zoom, save →



DINGO 2D

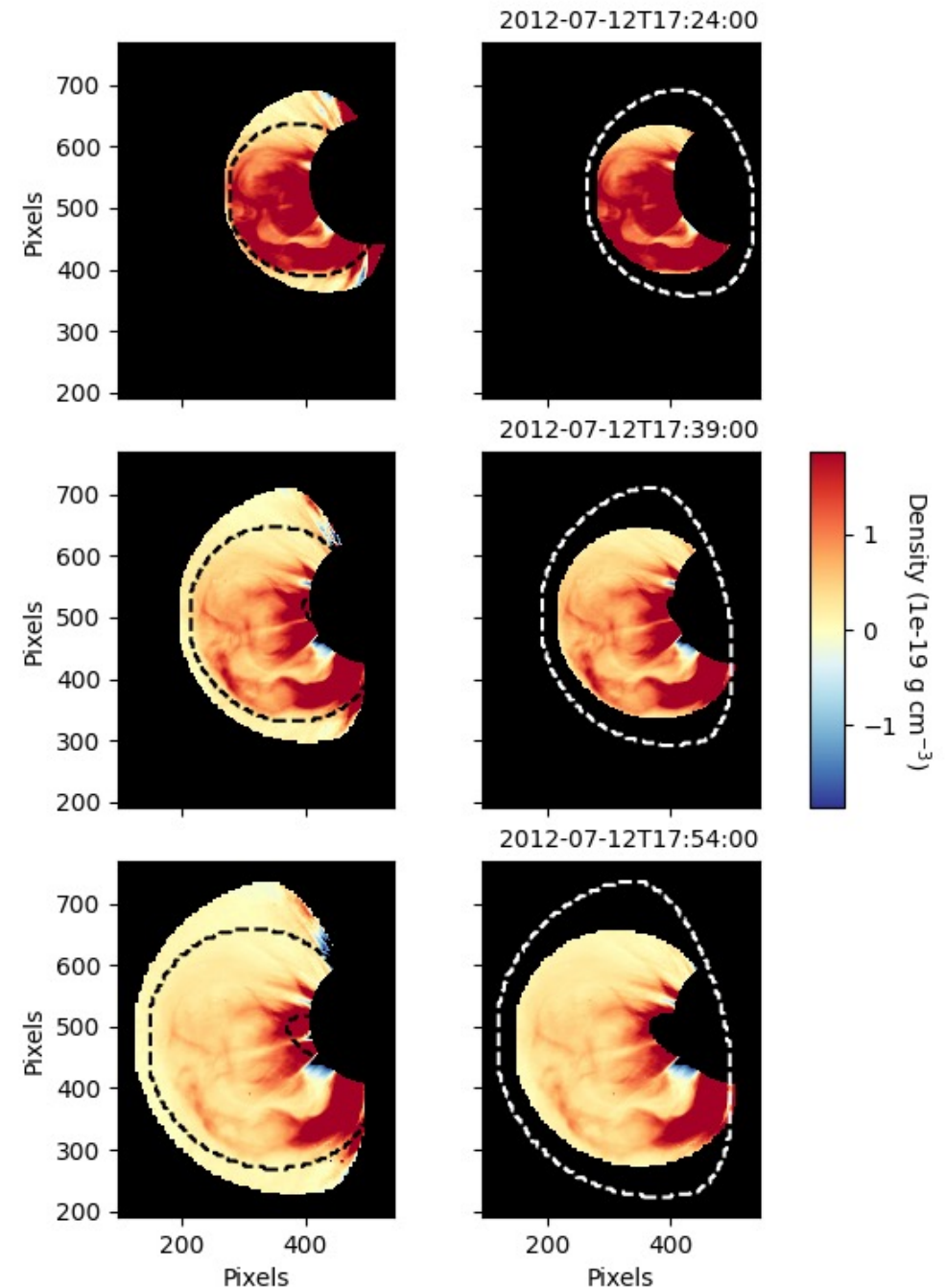
Outer+inner WFs shown as pairs.

*Black dashed line shows size of inner on outer panel
(white shows outer on inner)*

More than 4 times makes two columns of pairs

All panels use the same FoV and color bar.
logplot tag changes contour scaling to
logarithmic

DINGO will attempt to nicely size the figure
based on number of WFs and time steps and set
the contour levels. If this fails, the min_#/max_3
and figsize_#x# tags can be used to adjust the
aesthetics



DINGO 1D

Shows in place density (as seen in the remote observation) and in situ density

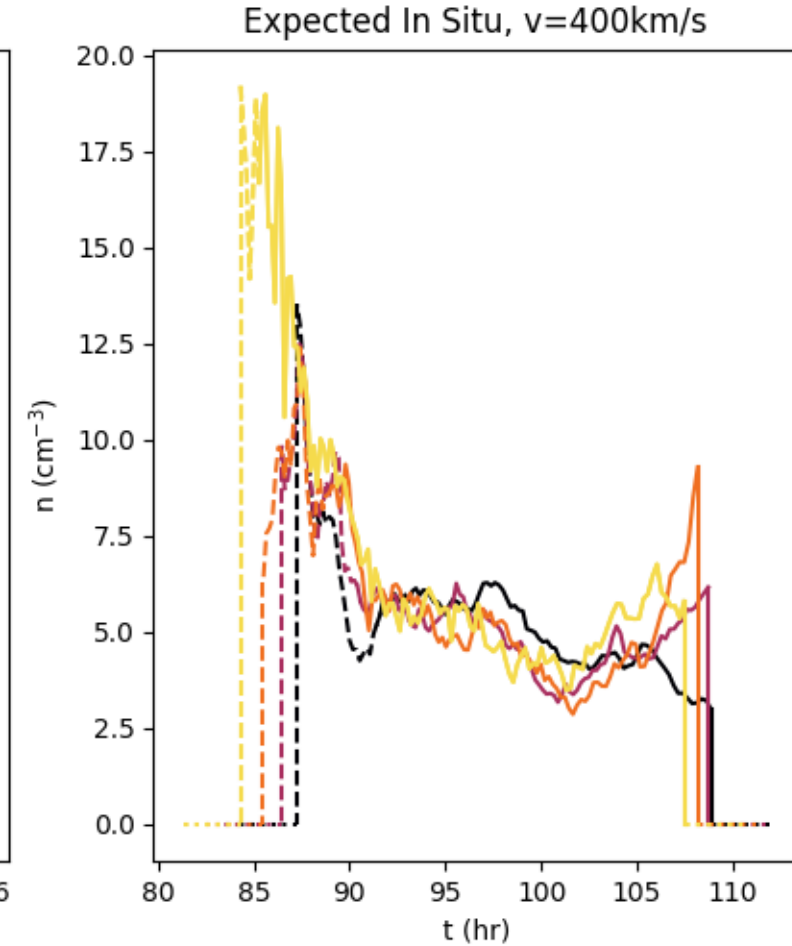
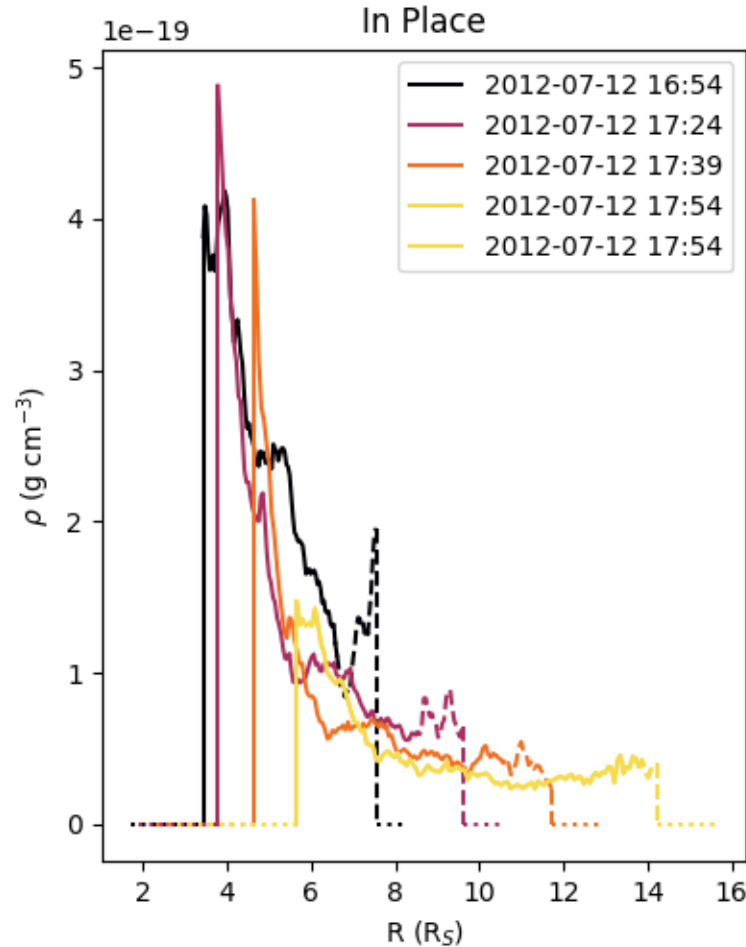
In place is along radial direction of in situ sat

Outer WF (sheath) shown as dashed and inner (ejecta) as solid.

Will also save data to files (dingo1d_IS and dingo1d_IP)

Very basic propagation model:

- Constant radial speed (*vcme*)
- Final nonradial size is some fraction of size expected from self similar expansion (*expf1*)
- Radial expansion speed is constant fraction of *vcme* (*expf2*)



Important Complications

For WOMBAT, logIDs can refer to multiple WF shapes but must be for the same time. These WFs will be displayed for all instruments within the pickle regardless of the log line instrument.

For DINGO, logIDs can refer to one or two WF shapes and multiple times but must be the same instrument and pickle. 3D DINGO only supports a single time and 2D has a max of eight times.

DINGO will warn about large PoS separations and cap the deprojection at a factor of 10. If it's a small percentage of points, particularly for WFs close to the Sun, this is fine. If the percentage is large the deprojection shouldn't be trusted

DINGO 1D is not an arrival time forecast model. Do not use it as such and expect very accurate results

Calculating masses for wide field images is not yet fully understood (see the WOMBAT paper). Use at your own risk!!!